Shape and Size Effects of Glass Mini-Channels on Infrared Sensors in Air–Water Two-Phase Flow

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Nomenclature

r	Reflection coefficient
t	Transmittance coefficient
n_1, n_2	Refractive index of first medium and second medium
I	Intensity of rays coming out of the medium
I_0	Intensity of incident ray
x	Photon path
q	Ray position
k	Wave vector

Greek Symbols

α	Absorption coefficient of the medium
θ_t, θ_1	Angle of transmittance, incidence
ω	Angular frequency

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1 Introduction

Simultaneous flow of two different phases of same fluid or different fluids is called two-phase flow. Gas–liquid two-phase flow occurs in bubble column reactors, microfluidic systems, electronic chip cooling, atomization in combustion processes, droplet formation, impaction in material processing, coating, mini heat exchanger tubes, and in power plants.

Void fraction plays a significant role in determining flow boiling heat transfer characteristics in mini/micro-channels. The effectiveness of compact heat exchangers with mini-channels depends upon the two-phase flow regimes, pressure drop, and void fraction as detailed by Ide et al. [5]. These parameters also play a vital role in designing of evaporators and condensers. Most importantly the void fraction which depends upon the flow regimes is a significant parameter which often varies along the length of the test section. Development of sensors for measuring void fraction is still a challenging task. One of the reasons is that the interface between the two phases in a gas–liquid two-phase regime is composed of a compressible and incompressible phase and exhibits different flow behavior.

The effect of channel size on two-phase flow patterns and void fraction in microchannels was studied by Sur and Liu [16]. A wire mesh sensor to determine void fraction in the vertical pipeline was developed by Prasser [12]. The results were compared with gamma-ray densitometry measurements. The bubble size distributions were measured. Rocha et al. [14] developed a conduction probe sensor to measure the instantaneous signal generated by the capacitance probe. The signal could be used for measuring two-phase flow void fraction. Lawal [9] used electrical impedance method for measuring void fraction in a multiphase flow. The capacitance value was reduced, while solid phase was added to a two-phase gas/liquid flow. Rocha and Moreira [15] experimentally measured Taylor bubble velocity and length. A multielectrode impedance sensor was used in which a rotating electric field sweeping across the test section was used to measure the necessary parameters involved.

Barreto et al. [3] measured void fractions using resistive impedance sensor. The influence of liquid flow regime on flow pattern and pressure drop was detailed. Cartellier [4] used optical probes to measure void fractions employing piercing methods and found that the results were in good agreement with direct visualization. Juliá et al. [7] measured void fractions in a bubbly flow using an optical probe. It was found that for non-perpendicular piercing, probe inclination leads to the additional source of underestimation. Jagannathan et al. [8] utilized laser pointer as a monochromatic coherent source of light and employed refraction phenomenon. Using Fresnel reflectance and transmittance coefficients [13], two-phase flow regime identification and slug velocity measurement were done.

All materials emit radiation if their temperature is above absolute zero in the form of electromagnetic waves or photons. If the emitted wavelength ranges between 700 nm and 1 mm, then it is termed as infrared radiation. When the temperature of a solid body increases, radiation of shorter wavelength is emitted. When ferrous materials are heated to a very high temperature, incandescence is in the visible region and starts to glow in red color [6]. Identification of two-phase flow regime and void fraction can be examined by a source which emits thermal radiation. Infrared transmitter and reflector is such a source which employs refraction phenomenon. The photons from IR LED (transmitter) when excited get scattered. But most of them get collimated as the construction of LED is such that it has a lens at its front end. The receiving end also has the same construction. The amount of scattered photon rays may be neglected in the absence of any medium. Adhavan et al. [1] determined the size and velocity of a slug using IR sensors without any complex filtering or processing circuit and validated with high-speed photography. Arunkumar et al. [2] identified the two-phase flow regime using infrared sensor and compared with CFD results employing a volume of fluids method. The results were in good agreement with high-speed photography. Recently, Mithran and Venkatesan [10] analyzed the effect of IR transceiver orientation on gas/liquid two-phase flow. It was noticed that the IR sensors were sensitive to angles at which the transceiver is placed. The thickness and shape of tubes varied the signal output which was not reported in the literature for multiphase flows.

Above literature shows that limited works are reported on using IR sensors for two-phase flow. When compared to other sensor arrangements, which require a separate signal processing circuit, IR sensors are simple in construction. The transceiver does not need any complicated signal processing techniques. The objective of the present work is to analyze the effect of wall thickness and shape of the channel on irradiation behavior of the infrared sensor during two-phase flow. The effect of various dimensional ratios and shapes (circular, square, and triangular) of test sections on the signal output characteristics of IR rays is analyzed for bubble and slug flow regimes. A numerical model is developed using COMSOL package to predict the irradiation behavior during single-phase flow. An IR transceiver circuit with the aid of NI-DAQ is used for measuring the intensity of scattered infrared rays in a gas/liquid two-phase flow. The results obtained are related with image processing techniques with good agreement. The developed numerical model will be useful to predict irradiation behavior of IR sensors for various shapes and sizes. The developed sensor may be further used along with arrays to determine volumetric void fraction.

2 Experimental Setup

The experimental setup for the present work is shown in Fig. 1. The setup is arranged in such a way that both IR sensing and imaging using high-speed camera is done simultaneously. Experiments are done in borosilicate glass test sections of internal diameters 2.50, 3.50, 5.20, and a 2 mm side square channel. The wall thickness of the test sections mentioned above is 0.3, 0.6, 0.6, and 0.5 mm, respectively.

Two separate E-spin SPK101 single channel syringe pumps with flow rate capacity of 0.4 nl/min–13.6 ml/min are used for air and water supply. Flow rates are controlled using a digital controller which is an integral part of the syringe pump. Water with

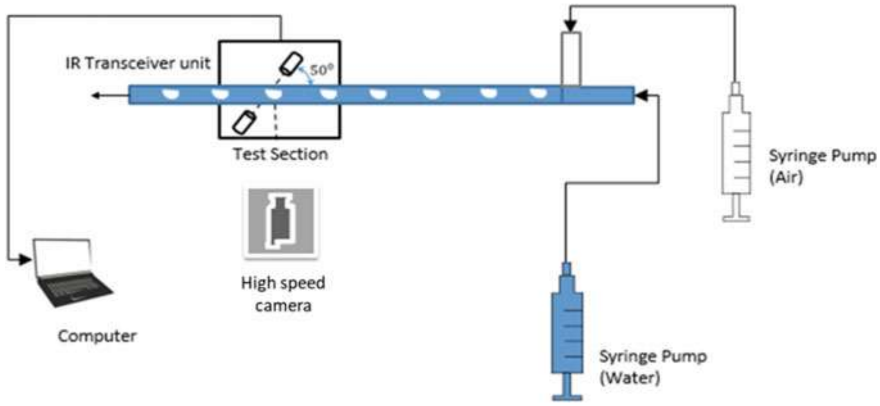


Fig. 1 Schematic diagram of the experimental setup

TDS concentration of 250 ppm is used for the experiments. The temperature of water used is 29 °C. Air and water are mixed in a T-section which is an integral part of the test section and has a vertical length of 5 cm. The IR transceiver circuit is positioned at a distance of 5 cm away from the mixing T-section so that the regime is fully developed. The IR transmitter and receiver face each other close to the tube in a straight line horizontally at an angle of 50° with the vertical. The angle chosen is based on the experiments initially conducted for single phase fluid so that maximum rays from the transmitter fall on the receiver unit. Two-phase flow is from right to left concerning the incident rays of the sensor. The length of the test section is 150 mm (uncertainty for length of the test section is in the order of $\pm$ 0.6 mm for all the tubes). BASLER CMOS acA2000 monochrome high-speed camera which can record at 340 fps at 2 MP resolution is positioned next to the sensor. Navitar zoom 7000 lens is used with the camera. The high-speed video and the sensor data are simultaneously recorded. IR transceiver signal is recorded using an NI-DAQ 9174 chassis with NI 9203 module with a maximum sampling rate of 200 kS/s using LabVIEW software. The image obtained is processed using MATLAB image processing technique which detects the edges of the flow regime and measures the dimensions of the flow regime.

3 Design and Simulation

Infrared rays are invisible to the naked eye at ambient temperature. It is electromagnetic radiation with a wavelength of the order of more than 700 nm. IR rays with longer wavelength are less scattered than visible light and are not obscured by gas or dust particles. All objects emit infrared radiation at all temperatures except at absolute temperature. The amount of radiation emitted is inversely proportional to the temperature of the object. The IR sensor used is a commercially available IR333A model. The IR transmitter and receiver are 5 mm in diameter and 7.6 mm length. IR

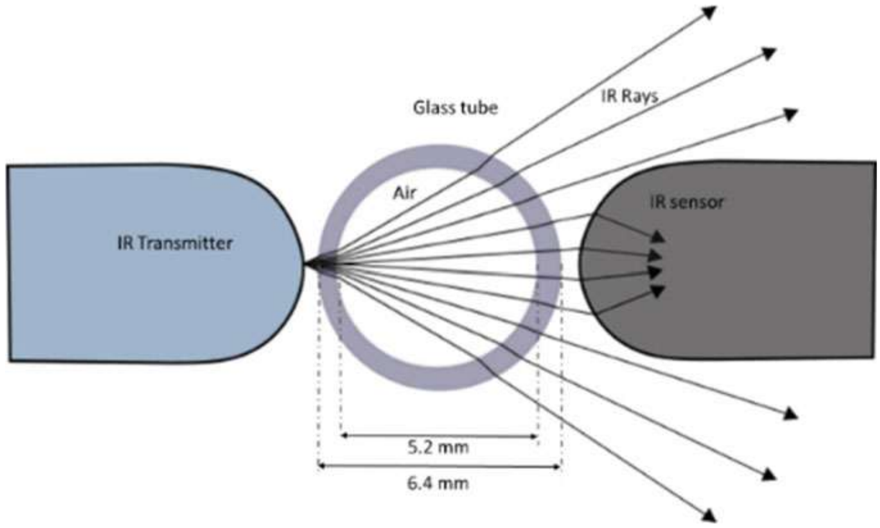


Fig. 2 IR rays for 5.2 mm diameter tube with 0.6 mm thickness (only air)

rays emitted from IR diode form a 30° conical shape at the exit of the transmitter, and after making contact with the glass tube, the rays diverge as shown in Fig. 2.

Focused conical rays emitted from IR diode are passed through the test section made of borosilicate glass. Initially, the intensity of IR rays is measured when only air is present inside the tube. The refractive indices of air, glass, and water are 1, 1.517, and 1.33, respectively. As the refractive index ratio for glass-air is 1.517 and for glass-water is 1.14, based on Snell's law, IR rays get refracted more while passing from glass to air than glass to water. The intensity of rays after divergence from the glass tube is relatively less, and only a small part of the incident IR rays reaches the receiver.

When the experiment is done with the tube filled with water, the diverging rays start to converge. A large part of incident IR ray reaches the receiver as shown in Fig. 3. As shown in Fig. 4, when a gas slug enters the test section, due to changes in the refractive index, the converged rays toward the sensor gets deflected. This will result in a reduction of measured current in IR receiver. Snell's law of refraction is used for depicting the above phenomena which are as follows

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \tag{1}$$

As the rays from IR transmitter are projected at 30° angles, the ray will meet at an incidence angle of θ_1 when compared with a normal tangent drawn on the circular boundary of the test section. Snell's law can be used to determine the angle of refraction θ_2. Similarly, all other angles of refraction can be calculated by considering the angle of refraction as the angle of incidence for other medium change.

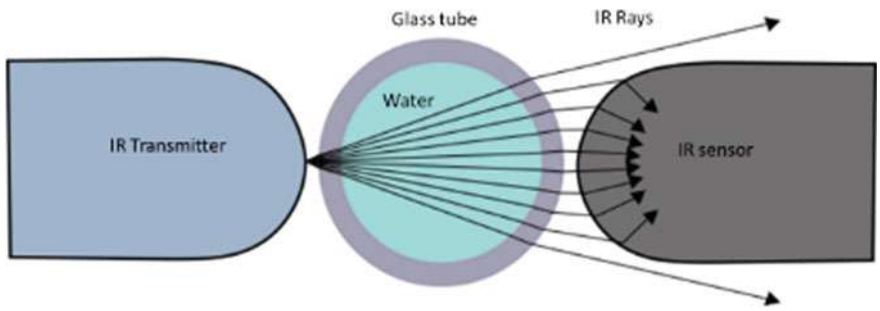


Fig. 3 IR rays for 5.2 mm diameter tube with 0.6 mm thickness—only water in the test section

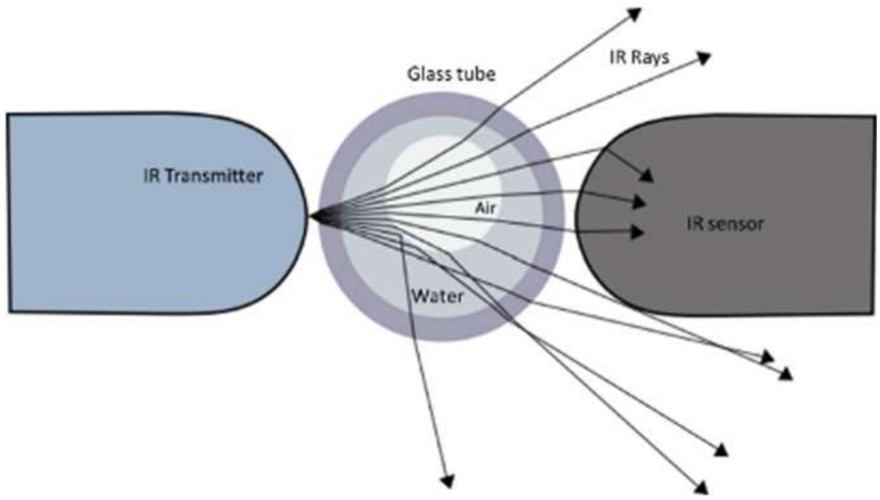


Fig. 4 IR rays for 5.2 mm diameter tube with 0.6 mm thickness—water and air in the test section

3.1 Method of Measurement Using IR Sensor

The limited region of the infrared wavelength spectrum with specific bandwidth is sensed by commercially available sensors. In the present work, the intensity of IR rays from IR transmitter is adjusted using a 10 k variable resistor from 0 to 3 V in IR transmitter circuit. A constant DC source of 5 V is used for power supply. The IR receiver is directly connected to an NI-cDAQ 9174 chassis with NI 9203 module (mA measurement) with its negative terminal as input with a 5 k variable resistor. The current variation in the sensor is measured using NI-cDAQ with 50 samples at 100 Hz for varying two-phase flow regimes of interest. The module is capable of measuring—20 to 20 mA. High-speed video image is instantaneously captured for the same flow regime. The images are captured using a high-speed camera at 300 frames per second. The captured image is processed by image processing techniques described in [11]. After the image is processed, the Euclidian pixel length is obtained. The size of the test section is already known, and hence, width and length of the bubbles are calculated.

3.2 Modeling in COMSOL Multiphysics

COMSOL Multiphysics software with ray optics module is used to solve the first order ordinary differential equations of Snells's law. The current generated from IR photodiode is related to a total number of photons entering the IR receiver Mithran and Venkatesan [10]. The governing equations for predicting ray direction in COMSOL ray optics are

$$\frac{\mathrm{d}q}{\mathrm{d}t} = \frac{\partial \omega}{\partial k} \tag{2}$$

The numerical model is developed with the test section, IR LED, and photodiode of dimensions as same as that of the experiment. The IR transceiver setup is positioned at an angle of 50° with respect to the test section for three different cross-sectional channels as shown in Fig. 5.

Horizontal circular tubes of internal diameter 2.50, 3.50, and 5.20 mm and thickness 0.3, 0.6, 0.6, and 0.5 mm, square channel of cross section—2 × 2 mm with 0.5 mm thickness, and an equilateral triangular test section of side 3.04 mm with 0.50 mm thickness is modeled for single phase water and air flow separately. The rays transmitted from LED incident on IR photodiode through a borosilicate glass test section and water with refractive index 1.517 and 1.33, respectively, are modeled using COMSOL package. The scale of IR rays is adjustable in terms of a number of rays. In the present case, it is chosen as 50 to have better visibility of the ray. Figure 5 shows the irradiation behavior of IR rays for various shapes of the test section. It can be observed from Fig. 5 that the divergence of IR rays in a triangular channel is more and fewer photons reaches the IR photodiode when compared to circular or square test sections. Due to the shape of the triangular test section, three different orientations with the IR transceiver are numerically studied as shown in Fig. 6. It is observed from the simulations that the percentage of rays reaching the IR photodiode is more in case 1 compared to other cases. In case 2 and case 3, the percentage of rays received for single phase air in test section is relatively higher, and percentage of rays received for a single phase of water in test section is less. While, in case 1, square channel and circular tubes the percentage of rays received for single phase water in test section is higher. The variation is observed as the divergence of IR rays has increased in case 2 and case 3 as transmitted photon rays are directed toward the corner of the triangular channel.

Hence, the orientation of triangular test section in case 1 is preferred as the divergence patterns of IR rays in this orientation is similar to the test sections of other

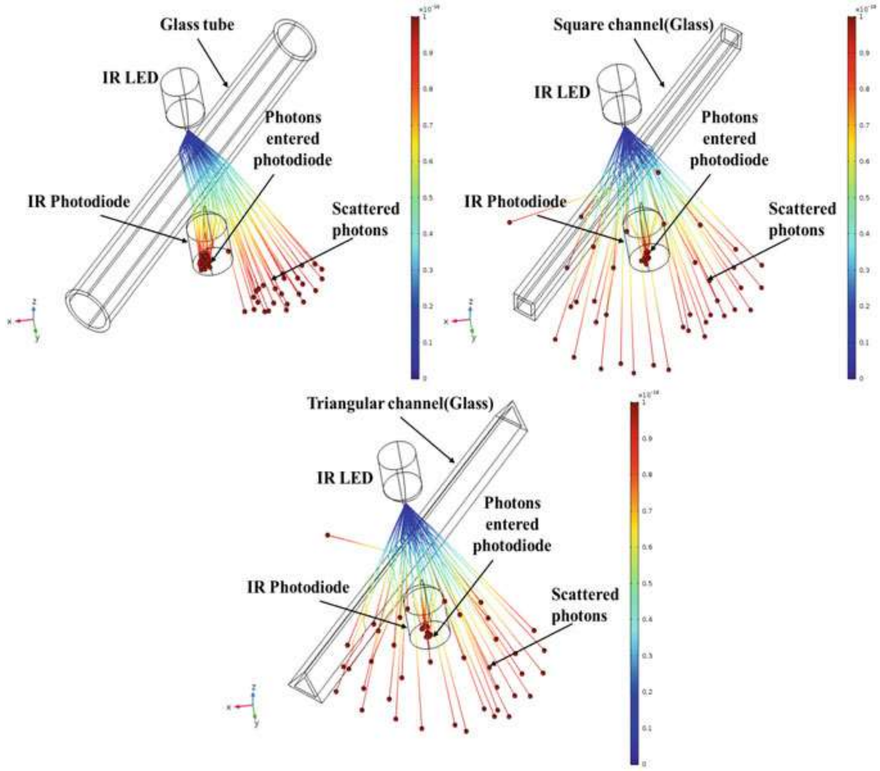


Fig. 5 Simulated COMSOL model for test sections of different shapes with IR transceiver setup

cross-sectional shapes. The other orientations of the square channel are not considered as the same phenomena would be observed, and there will not be any difference in signal output because of orientation. Two-phase flow experiments are planned on triangular channels separately for varying thickness and are scope for future work.

4 Results and Discussions

The various flow regimes visualized using high-speed photography and the corresponding current–time output obtained using IR sensor is compared for the square and circular test section. The experiments show that the cross-sectional void fraction (ratio of air to water at that cross-sectional area) is directly proportional to the amplitude difference of the current measured by the sensor. Initially, the experimental variation of current values is measured for single-phase flow of air and water on the same test section separately. The numerical results obtained from the model

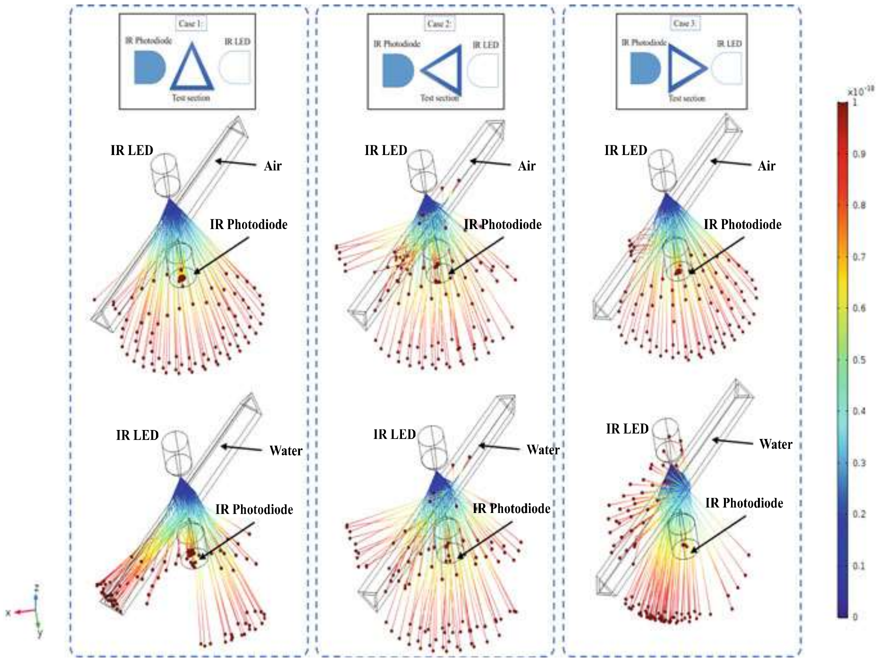
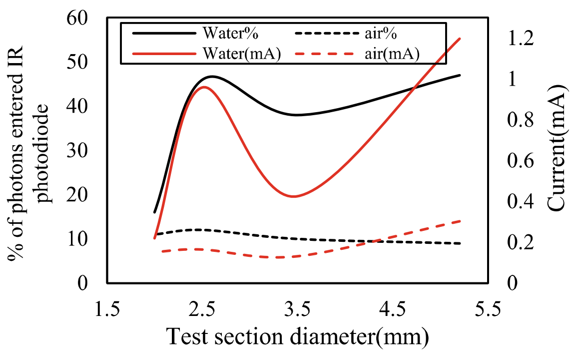


Fig. 6 Simulated COMSOL model for different orientation of triangular test sections with IR transceiver setup

developed using COMSOL package as in previous section is compared against the experiments and is shown in Fig. 7.

The current measured experimentally and the predicted percentage of photons numerically using COMSOL Multiphysics package is compared. From Fig. 7, it can be inferred that the measured mA value is lower for air and higher for water. The value of current measured with water for 2.5 mm tube is higher as the wall thickness of the tube is less than other test sections leading to increasing in the

Fig. 7 Current measured for air and water in different circular test sections



convergence of IR rays. The dimensional ratio (internal diameter to thickness) of test sections with inlet diameter 2.5 mm, 3.5 mm, and 5.2 mm are 0.12, 0.18, and 0.11, respectively. As the dimensional ratio value is similar for 2.5 and 5.2 mm test sections, the amplitude difference also appears to be the same for both the test sections. The amplitude difference measured for a 2 mm square channel is lower when compared to other circular test sections. The numerical results are also plotted in the same graph. The graph is plotted such that the current measured and percentage of photons of similar scale lie in y-axis for the same diameter of test section in x-axis. The patterns formed are similar for experiments as well as numerical data's, but there is a slight deviation of numerical results from experimental results. The difference is observed because the exact positioning of the transceiver with the test section could not be maintained similar to that of numerical model. It is also observed during experiments that the distance between IR LED, IR photodiode and test section also affects the IR irradiation behavior. The same phenomena are observed in numerical simulation. The amplitude difference measured for a 2 mm square channel is lower when compared to other circular test sections. Hence, only one square test section is compared to show the difference of signal pattern observed for the shape of the regime. Further, studies are required for bubble and slug regimes in various sizes of square channels and various shapes of the test section.

The values are shown in Fig. 7 and are taken as limits for two-phase flow regimes for the present work since the two-phase flow mixture will generate a signal whose current value will be between these limits. Bubbly and slug flow regimes are the two-phase flow regimes considered in the present work since accurate measurement of these regimes can be done with image processing. The bubble and slug regimes occurring in the four test sections are compared. High-speed photographs of bubble regimes for a 5.2 mm round tube and its processed image using the proposed algorithm are measured for superficial velocity of $U_g = 0.0116$ m/s and $U_l = 0.0081$ m/s. The superficial velocities are measured based on the data from syringe pump. The width and length of the bubbles in Fig. 8 are calculated as 4.54 and 10.92 mm using image processing technique. The signal output of the same bubbles measured using an IR transceiver is shown in Fig. 9. The variation in mA as a function of time is plotted to show the refracted IR rays. The direction of two-phase flow is from right to left. The measured mA as a function of time plotted represents the cross-sectional void fractions. The graph shows that when the size of the air void increases, drop in mA also increases. Due to the viscosity of the denser medium and velocity of the bubble, it can be seen that the front end of the void is sharper compared to its back end.

Fig. 8 Bubble in 5.2 mm tube

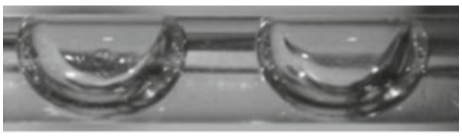
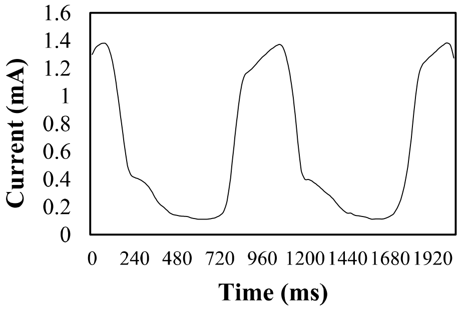


Fig. 9 IR measurements in 5.2 mm tube



The phenomena can also be observed from the plot where the drop of mA is stable and then drops further. The amplitude of current mA measured at the mid-section of the bubbles of size 4.59 mm and 4.54 mm are 0.112 mA and 0.114 mA, respectively. It may be recalled that the amplitude value for only water in the 5.2 mm inner diameter tube is 1.4 mA as detailed in Fig. 7. The difference in current value between only water and at mid-section of the bubble will give the current amplitude difference for cross-sectional void at the mid-section. The values recorded are 1.288 mA and 1.286 mA, respectively, which is proportional to the cross-sectional void fraction at the mid-section of the mentioned bubbles.

The observation is similar for regimes in a 3.5 mm tube and 2.50 mm tube with reduced amplitude of mA as shown in Figs. 10 and 11. The bubbles formed in this flow for given superficial velocities are almost identical in front and rear ends. Hence, the plot formed is mostly curved.

However, for 2 mm square section, the refraction of IR rays is different. It can be visualized from the plot that from the reference line (water only value of 1.70 mA), the value of current initially increases and then reduces which appears to be different from that of circular tubes. Further, the amplitude of mA is also reduced. The increase in mA is because of the flat section of the channel. In round tubes, the IR rays from the

Fig. 10 IR measurements in 3.5 mm tube

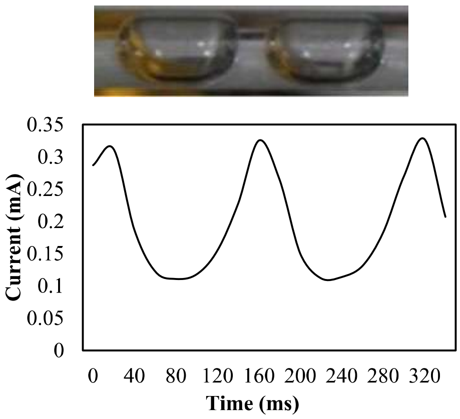
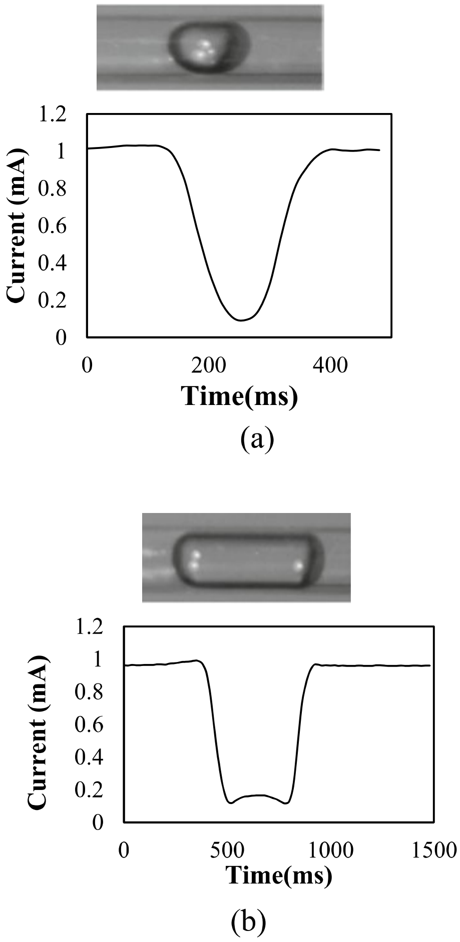


Fig. 11 IR measurements in 2.5 mm tube **a** for bubble and **b** for slug



transmitter are refracted more because of curvature shapes. In a square test section, since the section is flat, more photons reach the receiver without getting scattered. The surface of the glass is flat, and the refracted IR rays do not converge or diverge much and directly fall on IR receiver, which is shown in Fig. 12. A similar pattern is observed in slug regimes for the square test section.

The result obtained is verified for various sizes of bubble/slug using image processing technique. The determined width of the regime is compared with change in measured current amplitude difference. The difference is between the current output from IR sensor measured for only water in test section and current measured in the center of the bubble/slug as detailed earlier. Figure 13 shows the current amplitude difference for different width of the bubble and slug regimes in various test sections. The values shown in the plot are close to the current measured for only water in test

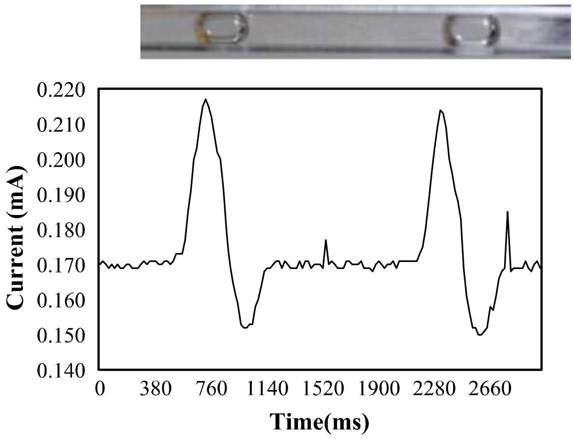


Fig. 12 IR measurements in 2 mm square tube

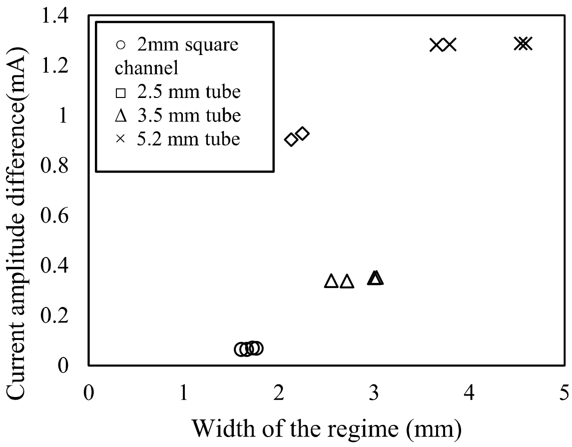


Fig. 13 Current amplitude difference for the width of the regimes in different test sections

section as shown in Fig. 6. Hence, the diametrical ratio of test section influences the irradiation behavior of IR rays.

The increase in current amplitude difference is observed with increase in width of the regime. For test sections, 2.5, 3.5, and 5.2 mm circular tubes, the amplitude difference values obtained are within the current range measured for only water and only air in the test section. However, the amplitude difference for 2 mm square test section is lower than the current measured for only air in the test section. The phenomena observed in the square channel is different from round tubes due to the curved shape of nose and tail of the regime and orientation of the IR transceiver setup, and hence, further analysis is required for varying thickness ratios of square channels and is scope of future work. The above results show that the thickness and size of the test sections play a vital role in the output of IR sensor. Comparison of the square, round and triangular (numerical) channels indicates that the output of the sensor depends on the shape of the channel also.

5 Conclusions

In the present study, two-phase flow experiments are conducted in the small round, square and triangular (numerical) channels of varying diametrical ratios which are kept in horizontal position. IR irradiation behavior is studied numerically using COMSOL package for the different dimensional ratio (thickness/cross section) of the test section. High-speed camera and image processing algorithm are used to determine the size of the bubble and slug flow regimes. The IR transceiver circuit with the help of NI-cDAQ is used for measuring the amount of scattering of IR rays for the same flow regimes. The intensity of scattering of IR rays is directly proportional to the size of the bubble/slug but influenced by the dimensional ratio (thickness/cross section) of the test section. No filtering or amplification circuit is required in the present technique. The numerical model developed can be further used to design an array of IR sensors capable of detecting volumetric void fraction.

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